5.04 Guide to Shaping Cuttings, Embankments and Tunnel Mouths

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1 Overview of the Terrain System

Terrain editing is performed in the World Editor by clicking the Painting Tools icon on the top left which resembles a paintbrush. This document assumes a basic level of familiarity with the terrain tools.

1.1 The Terrain Grid System

The terrain system in Rail Simulator uses a grid of approximately 8x8 metres. Every 8 metres is a point where the terrain can raised or lowered. Using the Height tool with the brush size set to 10 (it can be set smaller but 10 is a satisfactory value when individual terrain points need to be selected as it gives a little leeway) you can grab a single point on the terrain and drag it up and down.

So whenever a terrain tool is used, it must abide by this grid system. Sometimes this grid system can work in the users favour and other times it won't, in which case more skill and patience is required.

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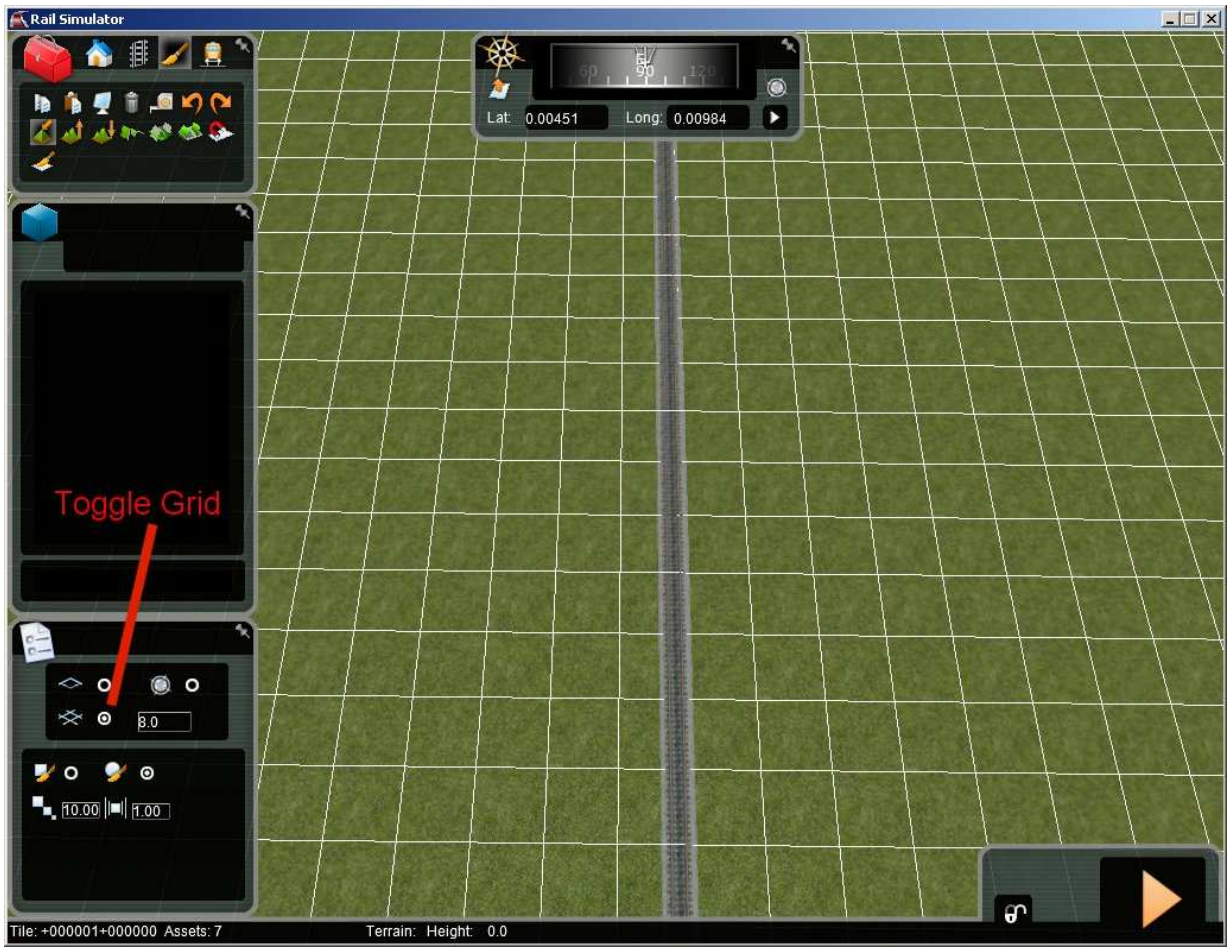
1.1.1 A Helpful Terrain Grid

In the context of forming cuttings and embankments it is very helpful if the terrain grid system follows the general direction of the track. You can determine this by clicking on the Grid button which has a default setting of 8 which matches the size of the terrain grid. This will overlay a grid on the terrain. Although this grid does not exactly show the location of each terrain point (as this grid is focussed on where the camera is located) it does show the direction of the terrain grid. Below is an example of the grid layout following the direction of the track, which is very helpful.



This is helpful because terrain deformation will occur in the direction of this grid. In the above picture the track is laid a few metres above the terrain. Using the Snap Tool you can deform the terrain so it snaps to the height of the track. You can see that because the track followed the direction of the terrain a smooth edge is created.

NOTE: A default rock texture is added to steep changes in terrain when you switch between edit and drive mode.

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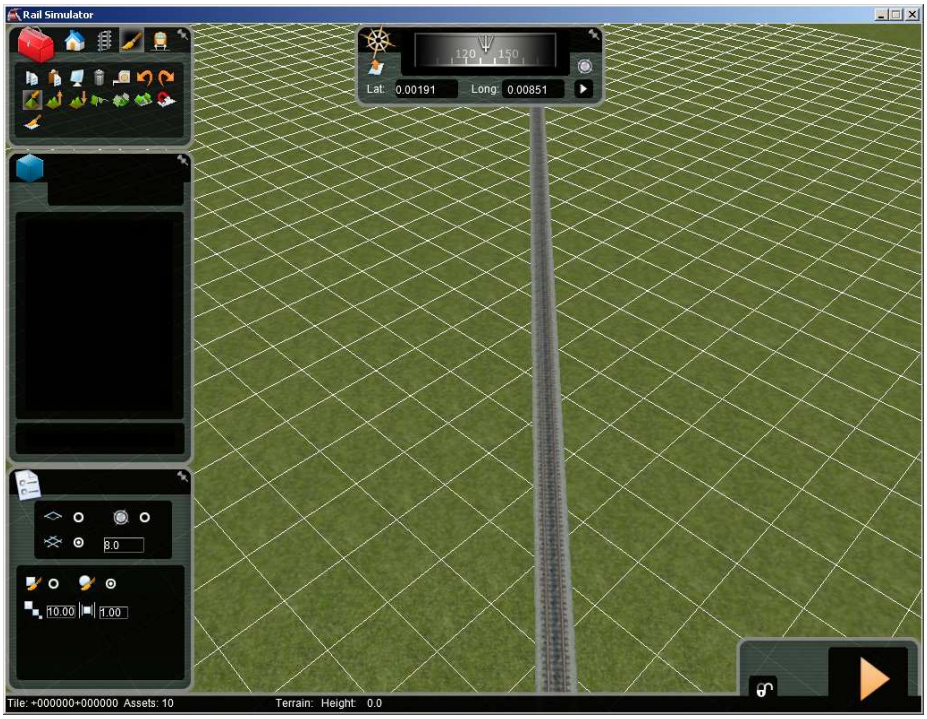
1.1.2 An Unhelpful Terrain Grid

If your track lies diagonally across the terrain grid then jagged edges will occur with the Snap Tool. Below is an example of a particularly unhelpful terrain grid.



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When the Snap Tool is used in this instance you can see a very different terrain effect.

This has created a saw-toothed effect. To create natural, smoother terrain some manual tweaking will be required after having applied the Snap Tool. How to do this and what settings to apply to the Snap tool to minimise this effect is described in the following sections.

In reality you will be faced with a mix of helpful and unhelpful terrain grid situations.

1.2 Using the Snap Terrain to Track Tool

The Snap Tool is used to deform the terrain so it snaps to the height of the track, or any other linear object within the tools radius. The snap tool is accessed via the icon resembling a magnet on the top left tab. Once clicked there are various settings which determine the effect of the tool. To use the tool, sweep it over the terrain like a brush.

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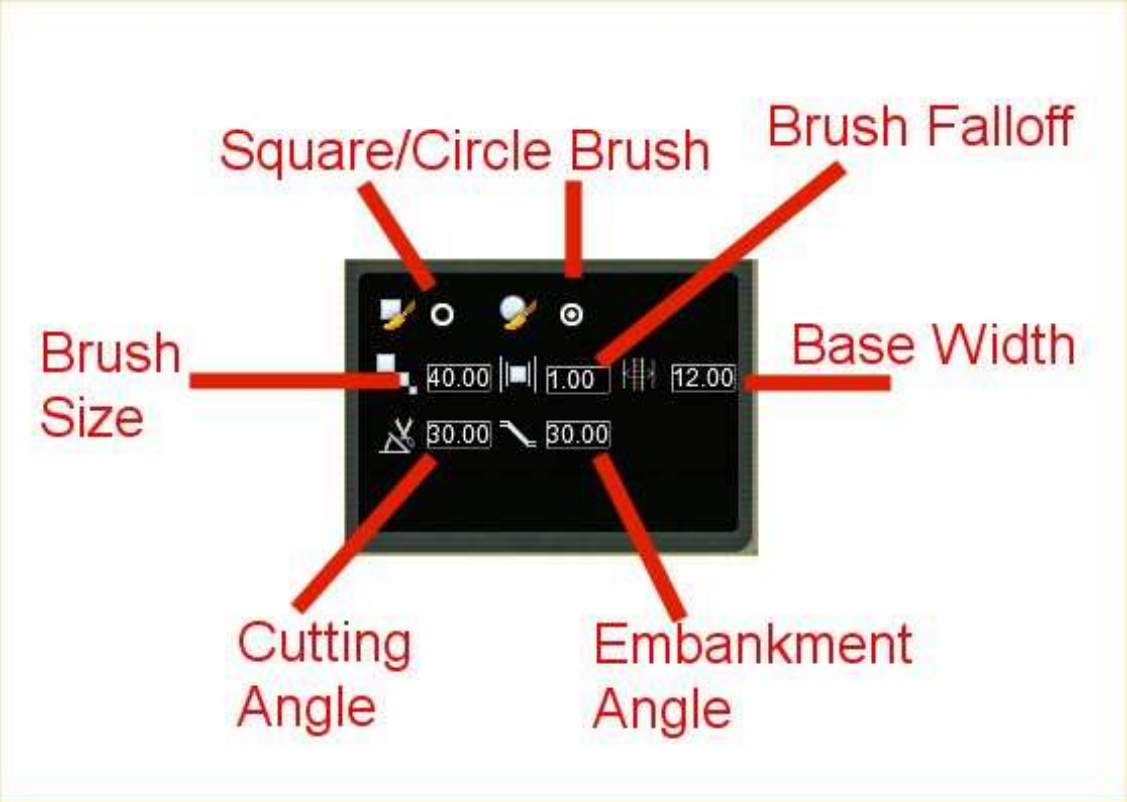

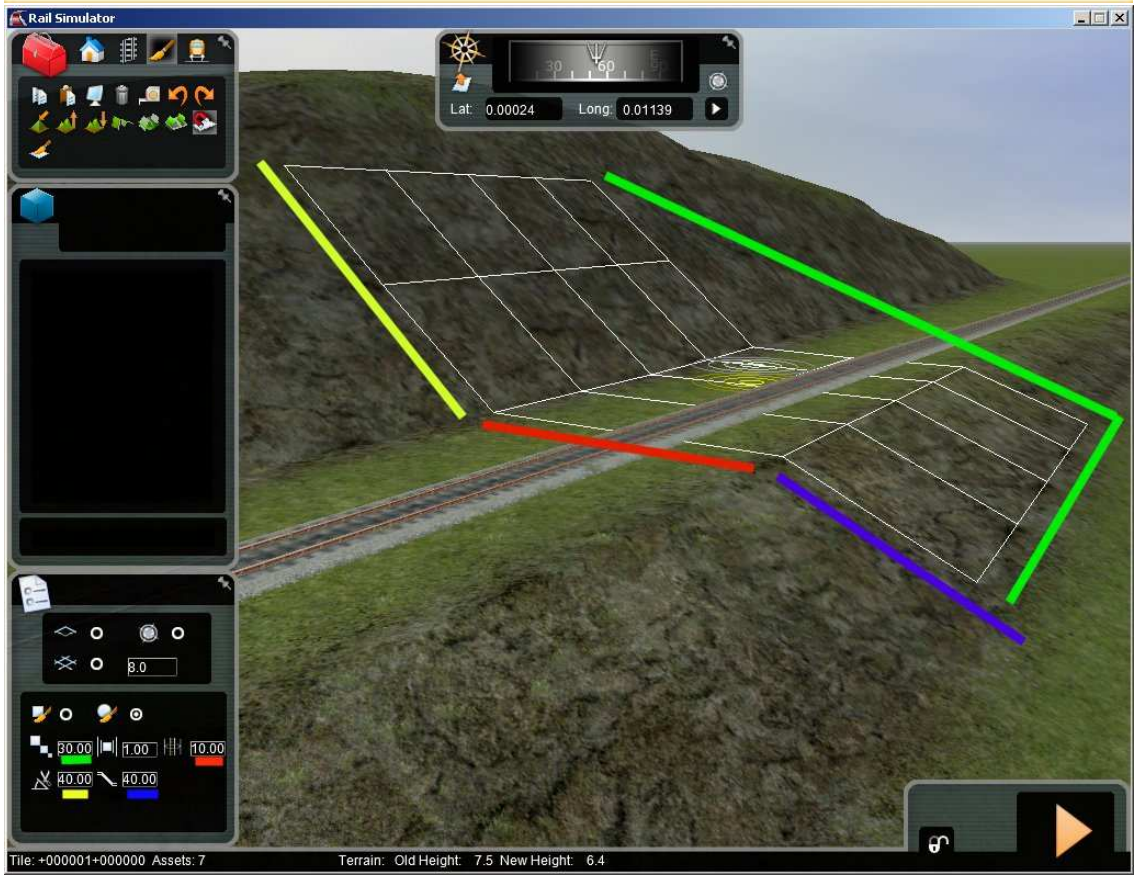

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In the context of cuttings and embankments, brush falloff has little effect so the default value is fine (you will notice the effect of falloff when raising or lowering large areas with a large brush size). Similarly, it doesn't have a massive effect whether a square or round brush is used.

The second image above shows where the base width (the "flat area" before the cutting and embankment values are applied), the cutting angle (affecting any terrain above track height), embankment angle (affecting any value below track height takes effect) is applied. The brush size is simply the size of the area affected by the settings.

1.3 Using the Height Terrain Tool for Detail

The Snap Tool is the main choice when creating cuttings and embankments but there will almost always be a need to tweak the terrain afterwards to get a better result, especially when jagged, saw-toothed edges are created.

The Height Tool is the best and most accurate tool for fixing such areas. The Height Tool is the first terrain tool icon and allows you to raise and lower terrain by moving the mouse up and down. Below is a saw-toothed embankment before tweaking.

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For fine tuning, set the brush size to 10. Left lick and hold a point on the terrain and move the mouse to raise or lower the jagged edges of the terrain to create a smooth embankment.

After tweaking with the Height Tool set to 10 the embankment is smooth and there are no jagged edges.

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2 Cutting and Embankment Examples

Although there is no "correct way" of creating an embankment or cutting, the following sections are intended to provide appropriate settings for creating typical examples. These settings can be used as a basis and adjusted until you get the result that best suits you.

2.1 A Gentle Cutting and Embankment

To create a gentle trackside embankments or cuttings such as in the images above, the following settings were used for the Snap Tool:

Brush Size: 40 (Too large and a performance hit is noticed as many terrain point alterations are calculated at once. This 40 brush can be swept to side to side along the track until no more terrain deformation is noticed).
Brush Falloff: 1 (Default).

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Base Width: 12
Cut Angle: 30 (Cuts at an angle of 30 degrees into terrain above track level after the 12 metre base width).
Embankment: 30 (Cuts at an angle of 30 degrees into terrain below track level after the 12 metre base width).

Single point terrain tweaking was then done using the Height Tool with a brush size of 10.

2.1.1 Cheating Tip 1: Templecombe Embankment

Sometimes the terrain system way not be able to create the short gentle slope you want; particularly if there is other track nearby. In the case of Templecombe on the Bath-Templecombe route there are two tracks close together at different heights and the terrain system was not producing satisfactory results. A linear object resembling a grass slope was created and placed instead of terrain. In the image below, the grass bank linear object (Grass Bank Loft under the Linear Objects filter in the editor) was laid offset from the two tracks on the right.

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2.2 A Steep Cutting and Embankment

Steeper trackside embankments or cuttings shown in the above images use the following approximate settings for the Snap Tool:

Brush Size: 40 (Too large and a performance hit is noticed as many terrain point alterations are calculated at once. This 40 brush can be swept to side to side along the track until no more terrain deformation is noticed).
Brush Falloff: 1 (Default).
Base Width: 12
Cut Angle: 60 (Cuts at an angle of 60 degrees into terrain above track level after the 12 metre base width).

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Embankment: 60 (Cuts at an angle of 60 degrees into terrain below track level after the 12 metre base width).

Again, single point terrain tweaking was then done using the Height Tool with a brush size of 10.

2.2.1 Cheating Tip 2: Cajon Pass Rocks

Steep rocky cuttings can be hard to create, especially if the terrain grid is working against you to create saw-tooth edges. On the Cajon Pass rock objects were embedded into the terrain where jagged edges would otherwise have been obvious.

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3 Overpasses and Underpasses

Overpasses and underpasses use much of the technique described above, but there are certain other factors to bear in mind:

- The Snap Tool does not distinguish between track, roads and other linear objects. When a track crosses a road it will try to adjust the terrain around both with unsatisfactory results for this task. Manual terrain editing will be required.
- Enough height needs to be allowed for your train to pass below any feature above the track, such as a road or rail overpass or a tunnel.
- Enough height needs to be allowed for a car to pass under the track for a road underpass, or a train for a rail underpass.
- Track and roads should never appear to be floating. There is almost always a way to prevent this, be it terrain manipulation or obfuscation with an object.

3.1 A Typical Example Overpass

The following workflow steps outline one method of creating an overpass such as the one shown in the image below:

1. Place the bridge

Before any terrain is jiggled with it is helpful to know the height to which you must move it to. There are several bridge assets in the game that will have clearance heights to suit your needs. Pick a bridge that you know has clearance room for your

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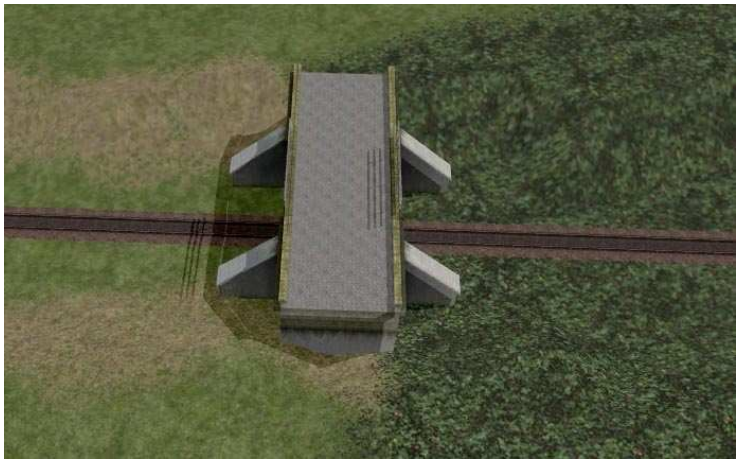
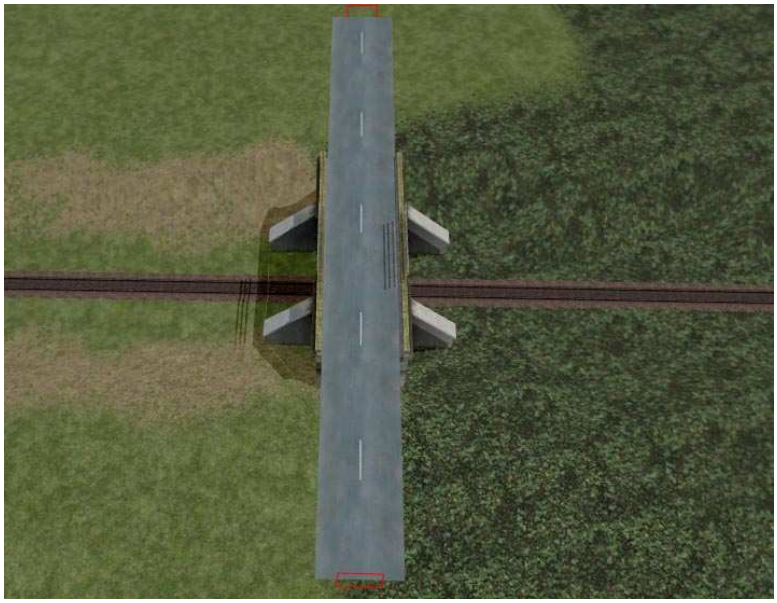
train to pass under and place it over the track. Make sure the bridge is not sunk or floating.



2. Place a suitable loft over the bridge shape

Most overpasses are for roads or track. In this example it is a road overpass. Place a straight section of the road linear object and move it so that it sits on top of the bridge shape.



3. Use the snap tool on the road ends

Once the linear object (in this case, a road) is in position over the bridge you can use the Snap Tool and sweep it over the ends, working towards the middle. There will be a point near the middle of the bridge where the Snap Tool will become ineffective as it can affect both the track and the newly placed, higher, linear object. Typical settings for the Snap Tool for this function are:

Brush Size: 30
Brush Falloff: 1 (Default).

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Base Width: 12
Cut Angle: 30 (Cuts at an angle of 30 degrees into terrain above track level after the 12 metre base width).
Embankment: 30 (Cuts at an angle of 30 degrees into terrain below track level after the 12 metre base width).

Entering drive mode will render the terrain and produce the correct lighting and shading for your new terrain. Re-enter the editor and the effect should look like the above image.

4. Tweak the terrain with a small brush
Although much of the terrain around the overpass is correct there may well be terrain points overlapping the track, or gaps under the road. Most of this can be fixed using the Height Tool with a brush size set to 10. You can pull individual terrain points up and down until you get a result you are happy with.

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5. Extend linear objects and use gradient points to angle them

Now that the terrain has been manipulated around the bridge you can extent the road as far as you want. Insert gradient points and pull them up and down if you want a sloping road. As long you do not sweep the Snap Tool too close to the terrain you have already deformed you can use the Snap Tool on the new parts of the road.

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Cheating Tip 3: Bridge Near Writhlington

In this example on the Bath-Templecombe route, the terrain on the left side of the bridge would either poke above the track or leave a big hole under the road.

An extra bridge end piece was seamlessly slotted next to it, doubling the depth of the brickwork on that side and covering up any terrain errors. On the right hand side there was a very small gap under the road, which was hidden with foliage.

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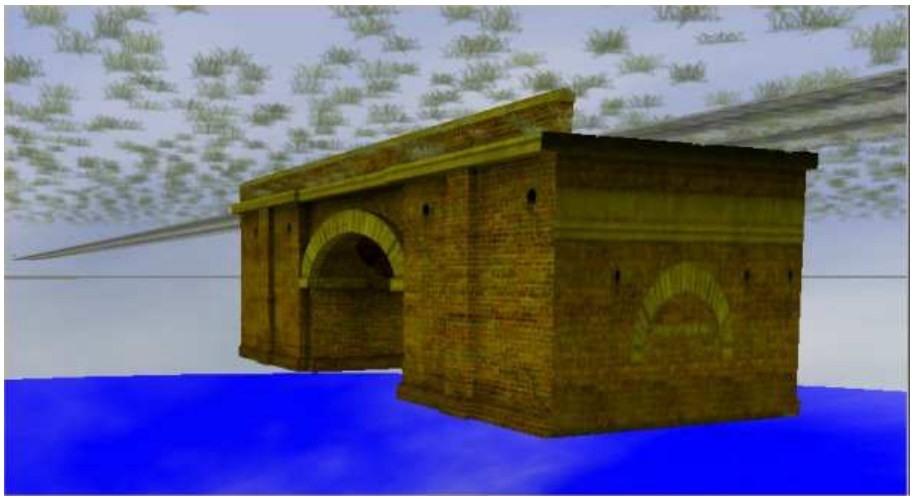
3.2 A Typical Example Underpass

Creating an underpass is much the same as creating an overpass. Suggested steps are detailed below:

1. Place a bridge asset to determine clearance under track

Again, use a bridge asset that will have clearance height to suit your needs. Make sure the bridge is sunken so that the track is running along the top of the bridge.



2. Place a suitable loft under the bridge shape

Underpasses can be used for different purposes such as roads, track, pedestrians or rivers. Each of these underpass types would require different clearance heights under the bridge. This example is a road overpass. Place a straight section of the road linear object and move it so that it sits under the bridge shape.

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Even if the end result does not require a linear object to be in the game it is useful in the creation of the terrain and can be deleted at the end.

3. Use the snap tool on the road ends

Once the linear object has been placed under the bridge at the correct height you can use the Snap Tool and sweep it over the ends, working towards the middle. Typical settings for the Snap Tool for this function are:

Brush Size: 30
Brush Falloff: 1 (Default).
Base Width: 10 (This small base width means the cutting under the track is quite narrow. If terrain triangles appear over the track try increasing this to 12)
Cut Angle: 50 (Cuts at an angle of 50 degrees into terrain above track level after the 10 metre base width).
Embankment: 50 (Cuts at an angle of 50 degrees into terrain below track level after the 10 metre base width).

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4. Tweak the terrain with a small brush

Now most of the terrain will be deformed but there will be terrain points overlapping the road or jagged points which need smoothing out. Most of this can be fixed using the Height Tool with a brush size set to 10. You can pull individual terrain points up and down until you get a result you are happy with.

5. Extend linear objects and use gradient points to angle them

Now that the terrain has been manipulated around the road you can extent the road further. Insert gradient points and pull them up and down if you want a sloping road. As long you do not sweep the Snap Tool too close to the terrain you have already deformed you can use the Snap Tool on the new parts of the road.

Additional objects can be placed to cover up any areas terrain manipulation failed. In this example, to avoid floating track at each side of the underpass, additional bridge assets have been placed which just look like a continuation of the wall. Foliage or a grass bank loft or another asset would have produced similar results.

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3.2.1 Cheating Tip 3: Victorville Underpass

To avoid a steep road and a sharp terrain height shift between road and track, as soon as it passes through the underpass, the road has been looped. This means the road is longer so the incline can be gentler. By the time the road rejoins running parallel to the track they are at the same height, thus avoiding any terrain conflict.

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4 Tunnel Mouths

This section will describe the steps needed to create a good looking tunnel mouth and presumes that track laying is complete, complete with the correct tunnel track profile. If a non-tunnel type track profile is laid then the camera will not automatically switch to cab view and the dark tunnel walls will not be present.

4.1 Creating a Typical Tunnel Entrance

The image below shows the basic terrain that will form the entrance to the tunnel. You may have terrain extracted from satellite data or you may have manually created your terrain. The track has been laid to the correct gradient and inside the hill the track has a tunnel type profile.

1. Snap the terrain to the track until you see the tunnel

Using the technique described earlier in this document to create a cutting you can run the Snap Tool along the track until you see the start of the tunnel track. In the image below the following settings were used on the Snap Tool:

Brush Size: 30
Brush Falloff: 1 (Default).
Base Width: 12
Cut Angle: 50 (Cuts at an angle of 50 degrees into terrain above track level after the 12 metre base width).
Embankment: 50 (Cuts at an angle of 50 degrees into terrain below track level after the 12 metre base width). Note: The embankment setting is not applied here as there is no terrain lower than track level.

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This has created a steep edge in the terrain around the start of the tunnel. Enough of the tunnel track type is exposed so that you can see it for lining up the tunnel mouth shape, but not too much is exposed as this will create a gap behind the tunnel mouth shape when it is placed. Some fine terrain tweaking may be needed at this point. Make the Height Tool brush size 10 and raise and lower points as needed.

2. Place the tunnel mouth shape

There is a selection of tunnel mouth shapes that can be used. In this example we are using a single track width tunnel. Navigate to the shape in Object Tools and select the Bridge / Viaduct / Tunnel filter in the middle left tab and either scroll down the list until you see the tunnel assets grouped together or select any object in the list and press T to jump to the objects beginning with the letter T.

Place the tunnel mouth shape so that it lines up with the start of the tunnel track type. Make sure it is rotated correctly and see what it looks like both from outside and inside the tunnel. You can fine rotate the object by selecting it and pressing F or G.

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3. **Place the Tunnel Hole Decal**

Now that the terrain and main shape for the tunnel is in place you can place a texture over the tunnel mouth that will act as a blank space when in drive mode. It is like a cartoon comedy hole that when placed on the terrain cuts through it. This is called a Tunnel Hole Decal.

In the image below you can where it is located in the browser list. Decals are flat textures which cover the shape of the terrain. They can be resized with the corner and side handles as well as rotated. Once the decal is placed on the tunnel it will be hard, though not impossible, to select again because the selection of the track or tunnel mouth are in close proximity. Place the decal on the ground near the tunnel and approximately resize it to the size of the inside of the tunnel shape and at a suitable orientation (also shown in the picture below). When satisfied, select it and drag it over the tunnel mouth. In editor mode it will look like a black texture but when in drive mode it will become transparent.

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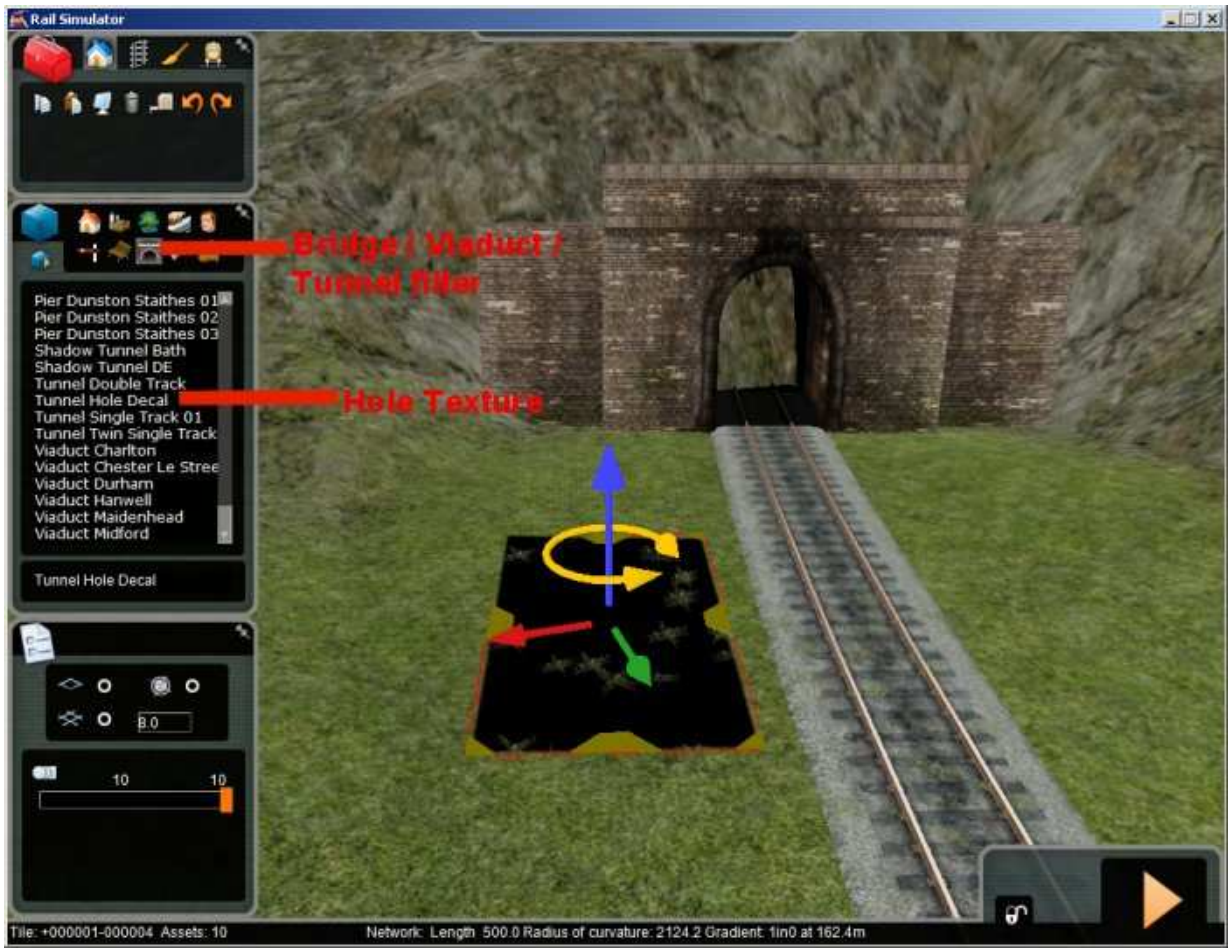

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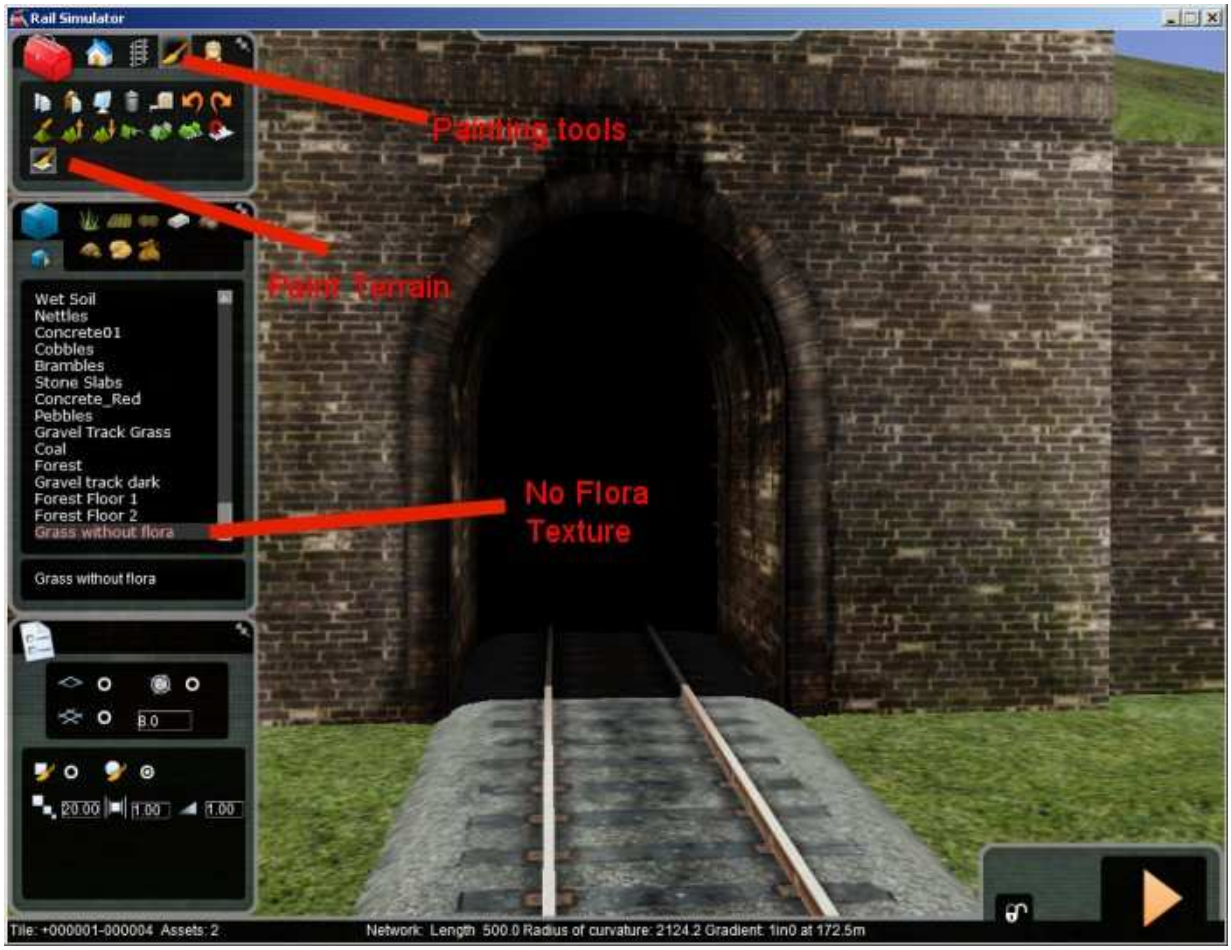
4. **Get rid of procedural flora over the tunnel hole**
If a texture is near the tunnel mouth which has procedural flora attached then you may get flora which appears floating in mid-air.

To avoid this, select the grass without flora texture from within the painting tools section as shown below.



Paint the immediate area with this texture until flora in the hole disappears. If some persists then you will need to paint a little further out.

5. Dress the tunnel mouth
With the major steps complete you can add the finishing touches to the tunnel area. Once you have a good looking tunnel mouth with the decal in place you should leave those few terrain points alone or you may affect how it sits on the terrain. Editing terrain a few points away from the tunnel is fine.

Minor errors such as any points where the tunnel track texture can be seen behind the tunnel shape can be mostly hidden with foliage or other shapes. A completed tunnel mouth is shown below.

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